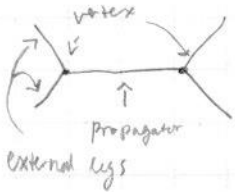


$$\mathcal{L} = \underbrace{\frac{1}{2} (\partial_\mu \phi_R)^2 - \frac{1}{2} m^2 \phi_R^2 - g \frac{\mu^{\epsilon/2}}{3!} \phi_R^3}_{= \mathcal{L}_R} + \underbrace{\frac{\xi_Z}{2} (\partial_\mu \phi_R)^2 - \frac{\delta_m}{2} \phi_R^2 - \frac{\delta_g \mu^{\epsilon/2}}{3!} \phi_R^3 + \delta_1 \phi_R}_{= \mathcal{L}_{CT}}$$

let's write: $p = \#$ of internal vertices, $n = \#$ of external lines, $v = \#$ of vertices.



Each external leg connects to one vertex and propagator to two vertices and a vertex to 2-external legs and a single propagator:

$\Rightarrow 3v = 2p + n$ must be satisfied

for example  has $v=2$, $n=4$, $p=1$

$$\rightarrow 6 = 2 \cdot 1 + 4 = 6 \quad \square \quad \text{and}$$

$L = p \cdot v + 1$ $\parallel$ each ^{internal} momentum integrated over, vertex includes one constraint on momentum conservation, overall momentum conservation gives +1

→ Each integral adds 6 powers of momentum to the numerator ($d=6-\epsilon$) (six dimensional) and every propagator adds 2 powers of momentum to denominator.

$\Rightarrow$ The degree of divergence is $D = 6L - 2p = 6 + \underbrace{4p - 6v}_{=-2n} = 6 - 2n$
 $\Rightarrow D = 6 - 2n$ means D is divergent for $n \leq 3$

$\Rightarrow$ 3-point function has $n=3 \Rightarrow D=0 \Rightarrow$ divergent
 $\Rightarrow$ 2-point function and 1-point function are also divergent since then we have $D \geq 0$

→ 1-, 2- and 3-point function are divergent.

We start with Z-point function:

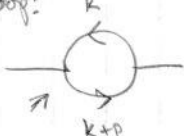
The two point function is  $= G(p^2) = \frac{i}{p^2 - m^2 - \Pi(p) + i\epsilon}$, using the geometric progression of the loops.

with $-i\Pi(p^2) = -i\underbrace{\Pi^{(1)}(p^2)}_{\text{1-loop correction}} + \underbrace{i(\delta_Z p^2 - \delta_m)}_{\text{The term from counter term Lagrangian via the Feynman rule.}}$

1-loop
Counter term:

 : $i(\delta_Z p^2 - \delta_m)$ is the two point function of counter term (Peskin page 325)

The 1-loop gets contribution only from the counter term above

loop:  : $-i\Pi^{(1)}(p^2) = (ig\mu^{\epsilon/2})^2 \frac{1}{2} \int \frac{d^d k}{(2\pi)^d} \frac{i}{(k^2 - m^2 + i\epsilon)} \frac{i}{((k+p)^2 - m^2 + i\epsilon)}$

Despite the arrows this is just a scalar propagator loop

The Feynman rule for the vertex

 is $-ig\mu^{\epsilon/2}$

Let's now calculate $-i\Pi^{(1)}(p^2)$

$$\begin{aligned} -i\Pi^{(1)}(p^2) &= -g^2\mu^\epsilon \frac{1}{2} \int \frac{d^d k}{(2\pi)^d} \frac{i^2}{(k^2 - m^2 + i\epsilon)} \frac{1}{((k+p)^2 - m^2 + i\epsilon)} \quad \parallel \text{Let's use Feynman parametrization} \\ &= +\frac{g^2\mu^\epsilon}{2} \int_0^1 dx \int \frac{d^d k}{(2\pi)^d} \frac{1}{(x[(k+p)^2 - m^2 + i\epsilon] + (1-x)[k^2 - m^2 + i\epsilon])^2} \end{aligned}$$

The denominator is D^2 such that

$$D = xk^2 + xp^2 - m^2x + i\epsilon x + k^2 - m^2 + i\epsilon - xk^2 + m^2x - i\epsilon x + x2k \cdot p$$

$$\Rightarrow D = k^2 + 2x \cdot k \cdot p + x^2 p^2 - x^2 p^2 + xp^2 - m^2 + i\epsilon$$

$$D = (k+xp)^2 - \underbrace{(x^2 p^2 - xp^2 + m^2 + i\epsilon)}_{\equiv \Delta}$$

Shift of variable: $l = k+xp \Rightarrow d^d k = d^d l$

$$\Rightarrow -i\Pi^{(1)}(p^2) = \frac{g^2\mu^\epsilon}{2} \int_0^1 dx \int \frac{d^d l}{(2\pi)^d} \frac{1}{(l^2 - \Delta)^2}$$

Let's now Wick rotate to Euclidean plane with $l^0 \rightarrow i l_E^0 \Rightarrow dl^0 \rightarrow i dl_E^0$
and $dl^i = dl_E^i \Rightarrow l^2 \rightarrow -l_E^2 \Rightarrow (l^2 - \Delta^2) \rightarrow (l_E^2 + \Delta)^2$

$$\Rightarrow -i\Pi^{(1)}(p^2) = i \frac{g^2\mu^\epsilon}{2} \int_0^1 dx \int \frac{d^d l_E}{(2\pi)^d} \frac{1}{(l_E^2 + \Delta)^2}$$

Peskin eq. (7.85)

gives us the integral :

$$\int \frac{d^d l_E}{(2\pi)^d} \frac{1}{(l_E^2 + \Delta)^2} = (4\pi)^{-d/2} \frac{\Gamma(2 - \frac{d}{2})}{\Gamma(2)} \Delta^{\frac{d}{2}-2} \parallel \Gamma(2) = 1$$

$$\frac{d}{2} = 6 - \epsilon$$

$$\Rightarrow -i\pi^{(1)}(p^2) = i g^2 \mu^\epsilon \frac{1}{2} \int_0^1 dx (4\pi)^{-3+\epsilon/2} \Gamma(\frac{\epsilon}{2}-1) \Delta^{1-\epsilon/2}$$

$$-i\pi^{(1)}(p^2) = i \frac{g^2}{2(4\pi)^3} \int_0^1 dx \left(\frac{4\pi\mu^2}{m^2}\right)^{\epsilon/2} \Gamma(\frac{\epsilon}{2}-1) \Delta \left(\frac{\Delta}{m^2}\right)^{-\epsilon/2} \quad \parallel \quad \text{We have added } 1 = \left[(m^2)^{\epsilon/2} \cdot (m^2)^{-\epsilon/2}\right]^{-1}$$

Let's now use the identity $\Gamma(\frac{\epsilon}{2}-1) = (\frac{\epsilon}{2}-1)^{-1} \Gamma(\frac{\epsilon}{2})$

And $a^{\epsilon/2} = e^{\frac{\epsilon}{2} \ln a} \approx 1 + \frac{\epsilon}{2} \ln(a) + \mathcal{O}(\epsilon^2)$ for small ϵ
 similarly $a^{-\epsilon/2} = e^{-\frac{\epsilon}{2} \ln(a)} \approx 1 - \frac{\epsilon}{2} \ln(a) + \mathcal{O}(\epsilon^2)$ for small ϵ

$$\Rightarrow \left(\frac{4\pi\mu^2}{m^2}\right)^{\epsilon/2} \approx 1 + \frac{\epsilon}{2} \ln\left[\frac{4\pi\mu^2}{m^2}\right] \quad \text{to the first order in } \epsilon$$

$$\Rightarrow \left(\frac{\Delta}{m^2}\right)^{-\epsilon/2} \approx 1 - \frac{\epsilon}{2} \ln\left[\frac{\Delta}{m^2}\right] \quad \text{to the first order in } \epsilon$$

Let's also use $(\frac{\epsilon}{2}-1)^{-1} = - (1 - \frac{\epsilon}{2})^{-1} \approx - (1 + \frac{\epsilon}{2})$
 and the identity $\Gamma(\frac{\epsilon}{2}) \approx \frac{2}{\epsilon} - \gamma_E + \mathcal{O}(\epsilon)$

$$\Rightarrow -i\pi^{(1)}(p^2) = -i \frac{g^2}{2(4\pi)^3} \int_0^1 dx \left(\frac{2}{\epsilon} - \gamma_E\right) \left(1 + \frac{\epsilon}{2}\right) \left(1 + \frac{\epsilon}{2} \ln\left(\frac{4\pi\mu^2}{m^2}\right)\right) \left(1 - \frac{\epsilon}{2} \ln\left(\frac{\Delta}{m^2}\right)\right) \Delta$$

ϵ is small $\Rightarrow \left(1 + \frac{\epsilon}{2} \ln\left(\frac{4\pi\mu^2}{m^2}\right)\right) \left(1 - \frac{\epsilon}{2} \ln\left(\frac{\Delta}{m^2}\right)\right) \approx 1 + \frac{\epsilon}{2} \ln\left(\frac{4\pi\mu^2}{m^2}\right) - \frac{\epsilon}{2} \ln\left(\frac{\Delta}{m^2}\right)$

and we will have $\epsilon \rightarrow 0$, there is only one $\frac{2}{\epsilon}$ term so we can take all ϵ^2 terms to go to zero and only spare terms $\sim \epsilon$ before multiplying by $\left(\frac{2}{\epsilon} - \gamma_E\right)$

$$\Rightarrow -i\pi^{(1)}(p^2) = -i \frac{g^2}{2(4\pi)^3} \int_0^1 dx \left(\frac{2}{\epsilon} - \gamma_E\right) \left(1 + \frac{\epsilon}{2} \ln\left(\frac{4\pi\mu^2}{m^2}\right) - \frac{\epsilon}{2} \ln\left(\frac{\Delta}{m^2}\right)\right) \Delta \, dx$$

Now we only spare terms that are of order $\frac{1}{\epsilon}$ or $\epsilon^0 = 1$ since $\epsilon \rightarrow 0$

$$\Rightarrow \left(\frac{2}{\epsilon} - \gamma_E\right) \left(1 + \frac{\epsilon}{2} \ln\left(\frac{4\pi\mu^2}{m^2}\right) - \frac{\epsilon}{2} \ln\left(\frac{\Delta}{m^2}\right)\right) \approx \frac{2}{\epsilon} - \gamma_E + \ln\left[\frac{4\pi\mu^2}{m^2}\right] - \ln\left[\frac{\Delta}{m^2}\right]$$

$$\Rightarrow -i\pi^{(1)}(p^2) = -i \frac{g^2}{2(4\pi)^3} \left[\int_0^1 dx \left(\frac{2}{\epsilon} - \gamma_E + \ln\left[\frac{4\pi\mu^2}{m^2}\right]\right) \Delta - m^2 \int_0^1 dx \frac{\Delta}{m^2} \ln\left(\frac{\Delta}{m^2}\right) \right]$$

$\Delta = m^2 - x(1-x)p^2 + i\epsilon \approx m^2 - x(1-x)p^2$ since $\epsilon \rightarrow 0$ (i ϵ -prescription was not necessary AFTER Wick rotation)
 The integral $\int_0^1 dx (m^2 - x(1-x)p^2)$ gives $m^2 - (\frac{1}{2} - \frac{1}{3})p^2 = m^2 - \frac{1}{6}p^2$

$$\Rightarrow -i\pi^{(1)}(p^2) = -i \frac{g^2}{2(4\pi)^3} \left[\left(m^2 - \frac{p^2}{6}\right) \left(\frac{2}{\epsilon} - \gamma_E + \ln\left[\frac{4\pi\mu^2}{m^2}\right]\right) - m^2 \int_0^1 dx \frac{\Delta}{m^2} \ln\left(\frac{\Delta}{m^2}\right) \right]$$

The divergent part is with $\frac{2}{\epsilon} \Rightarrow -i \frac{g^2}{2(4\pi)^3} \left(m^2 - \frac{p^2}{6}\right) \frac{2}{\epsilon}$ is the divergent part
 The counter terms are of form $i(\delta_Z p^2 - \delta m)$

Which means that in the minimum subtraction scheme we get

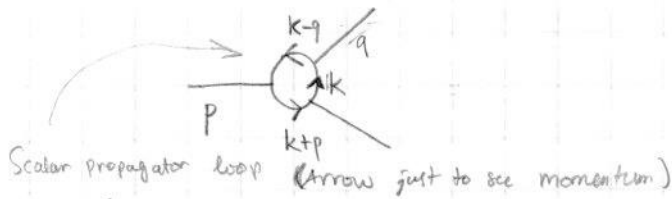
$$\delta m^{MS} = -\frac{g^2}{2} \frac{m^2}{(4\pi)^3} \frac{2}{\epsilon} \quad \text{and} \quad \delta z^{MS} = -\frac{g^2}{12(4\pi)^3} \frac{2}{\epsilon} \quad \square$$

Exercise 7.9

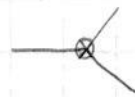
Now we tackle the 3-point function.

The 3-point function in the first order is , Now we introduce

the 1-loop correction to the vertex (The Feynman rule is $-ig\mu^{\epsilon/2}$ for the vertex)



with the counter term



$$= -i\delta g\mu^{\epsilon/2}$$

Feynman rule straight from the Lagrangian.

The loop integral is

$$\Gamma_3^{(1)}(p^2) = \Gamma^{(1)}(p^2) = \underbrace{(-ig\mu^{\epsilon/2})^3}_{\text{From 3 vertex along the loop}} \int \frac{d^d k}{(2\pi)^d} \frac{i}{(k+p)^2 - m^2 + i\epsilon} \frac{i}{k^2 - m^2 + i\epsilon} \frac{i}{(k-q)^2 - m^2 + i\epsilon}$$

From 3 vertex along the loop

Let's start by using Feynman parametrization

$$\frac{1}{ABC} = \int dx dy dz \frac{2! \delta(x+y+z-1)}{[xA+yB+zC]^3}$$

$$\Rightarrow \Gamma^{(1)}(p^2) = 2(-ig\mu^{\epsilon/2})^3 \int \frac{d^d k}{(2\pi)^d} \int dx dy dz \delta(x+y+z-1) [x(k+p)^2 + yk^2 + z(k-q)^2 + (x+y+z)(i\epsilon - m^2)]^{-3}$$

Integrate over y to get

$$\Gamma^{(1)}(p^2) = 2(-ig\mu^{\epsilon/2})^3 \int_0^1 dx \int_0^{1-x} dz \int \frac{d^d k}{(2\pi)^d} [x(k+p)^2 + (1-x-z)k^2 + z(k-q)^2 + i\epsilon - m^2]^{-3}$$

The denominator D^3 can be simplified further:

$$D = \underline{xk^2} + x p^2 + 2k p x + k^2 - \underline{xk^2} - \underline{zk^2} + \underline{zk^2} - 2k q z + z q^2 + i\epsilon - m^2$$

$$= k^2 + 2k p x - 2k q z + x p^2 + z q^2 + i\epsilon - m^2 = k^2 + 2k(p x - q z) + x p^2 + z q^2 + i\epsilon - m^2$$

$$= k^2 + 2k(p x - q z) + (p x - q z)^2 - (p x - q z)^2 + x p^2 + z q^2 + i\epsilon - m^2 \quad \parallel \quad \begin{aligned} p^2 x^2 + q^2 z^2 - 2x z p q \\ = (p x - q z)^2 \end{aligned}$$

$$= (k + p x - q z)^2 + x p^2 (1-x) + z q^2 (1-z) + 2x z p \cdot q - m^2 + i\epsilon$$

Let's do change of variable: $l = k + p x - q z \Rightarrow dk = dl$

$$\text{and} \quad \Delta = -x p^2 (1-x) - z q^2 (1-z) - 2x z p \cdot q - m^2$$

$$\Rightarrow D = l^2 - \Delta + i\epsilon \quad \Rightarrow \quad \Gamma^{(1)}(p^2) = 2(-ig\mu^{\epsilon/2})^3 \int_0^1 dx \int_0^{1-x} dz \int \frac{d^d l}{(2\pi)^d} \frac{1}{(l^2 - \Delta + i\epsilon)^3}$$

Now Wick rotate using $l^0 \rightarrow i l_E^0$ and $l^i \rightarrow l_E^i$

$\Rightarrow l^2 \rightarrow -l_E^2$ and $d^d l \rightarrow i d^d l_E$ and we drop the i since denominator $\neq 0$ anywhere in Euclidean space.

$$\Rightarrow \Gamma^{(1)}(p) = 2i(-ig\mu^{\epsilon/2})^3 \int_0^1 dx \int_0^{1-x} dz \int \frac{d^d l_E}{(2\pi)^d} \frac{-1}{(l_E^2 + \Delta)^3}$$

From Peskin eq. (7.85) : $\int \frac{d^d l_E}{(2\pi)^d} \frac{1}{(l_E^2 + \Delta)^n} = (2\pi)^{-d/2} \frac{\Gamma(n-d/2)}{\Gamma(n)} \Delta^{\frac{d}{2}-n}$

$$\Rightarrow \int \frac{d^d l_E}{(2\pi)^d} \frac{1}{(l_E^2 + \Delta)^3} = (4\pi)^{-\frac{d}{2}} \frac{\Gamma(3-d/2)}{\Gamma(3)} \Delta^{\frac{d}{2}-3}$$

In our case $d = 6 - \epsilon$

$$\Rightarrow \int \frac{d^d l_E}{(2\pi)^d} \frac{1}{(l_E^2 + \Delta)^3} \stackrel{d=6-\epsilon}{=} 4\pi^{-3+\epsilon/2} \frac{\Gamma(\frac{\epsilon}{2})}{\Gamma(3)} \Delta^{\frac{\epsilon}{2}}$$

$$\parallel \Gamma(3) = 2$$

$$\Rightarrow = \frac{1}{2} \frac{1}{(4\pi)^3} (4\pi)^{\epsilon/2} \Gamma(\epsilon/2) \Delta^{-\epsilon/2}$$

$$\Rightarrow \Gamma^{(1)}(p) = -2i(-ig\mu^{\epsilon/2})^3 \frac{(4\pi)^{\epsilon/2}}{2(4\pi)^3} \int_0^1 dx \int_0^{1-x} dz \Gamma(\epsilon/2) \Delta^{-\epsilon/2} \parallel (-i)^3 = -i \cdot (-i)^2 = i$$

$$\Delta^{-\epsilon/2} = \exp(-\frac{\epsilon}{2} \ln \Delta) \approx 1 - \frac{\epsilon}{2} \ln \Delta + \mathcal{O}(\epsilon^2)$$

$$\Gamma(\frac{\epsilon}{2}) \approx \frac{2}{\epsilon} - \gamma_E + \mathcal{O}(\epsilon)$$

$$(4\pi)^{\epsilon/2} \mu^{3\epsilon/2} = \mu^{\epsilon/2} (4\pi\mu^2)^{\epsilon/2} = \mu^{\epsilon/2} \exp(\frac{\epsilon}{2} \ln(4\pi\mu^2)) \approx \mu^{\epsilon/2} (1 + \frac{\epsilon}{2} \ln(4\pi\mu^2) + \mathcal{O}(\epsilon^2))$$

We spare the $\mu^{\epsilon/2}$ from approximation since the counter term is proportional to $\mu^{\epsilon/2}$

$$\Rightarrow \Gamma^{(1)}(p^2) = \frac{g^3 \mu^{\epsilon/2}}{(4\pi)^3} \int_0^1 dx \int_0^{1-x} dz (1 + \frac{\epsilon}{2} \ln(4\pi\mu^2) + \mathcal{O}(\epsilon^2)) (\frac{2}{\epsilon} - \gamma_E + \mathcal{O}(\epsilon)) (1 - \frac{\epsilon}{2} \ln \Delta + \mathcal{O}(\epsilon^2))$$

After multiplying we drop terms of order $\mathcal{O}(\epsilon)$ and higher since ϵ is small.

$$\Rightarrow \Gamma^{(1)}(p^2) \approx \frac{g^3 \mu^{\epsilon/2}}{(4\pi)^3} \int_0^1 dx \int_0^{1-x} dz \left[\frac{2}{\epsilon} - \gamma_E + \ln(4\pi\mu^2) - \ln(\Delta) \right]$$

$$\int_0^1 dx \int_0^{1-x} dz = \int_0^1 dx (1-x) = \left(x - x^2 \frac{1}{2} \right) \Big|_0^1 = 1 - \frac{1}{2} = \frac{1}{2}$$

$$\Gamma^{(1)}(p^2) = \frac{g^3 \mu^{\epsilon/2}}{2(4\pi)^3} \left(\underbrace{\frac{2}{\epsilon} - \gamma_E + \ln(4\pi\mu^2)}_{\text{the divergent part}} - \int_0^1 dx \int_0^{1-x} dz \ln(\Delta) \right)$$

The three-point function is $-i\Gamma(p^2) = -i\Gamma^{(1)}(p^2) - i\delta_g \mu^{\epsilon/2} - \underbrace{ig\mu^{\epsilon/2}}_{\Rightarrow ig \text{ as } \epsilon \rightarrow 0}$ so doesn't have divergence.

In the $\overline{MS}$ -scheme we recognise $\delta_g^{\overline{MS}}$ as being term that exactly cancels the divergent part of $\Gamma^{(1)}(p^2)$

The divergent part of $\Gamma^{(1)}(p^2)$ is $\mu^{\epsilon/2} \frac{g^3}{2(4\pi)^3} \frac{2}{\epsilon} \equiv \Gamma_{\text{div}}^{(1)}(p^2)$

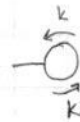
$$\Rightarrow -i \Gamma_{\text{div}}^{(1)}(p^2) - i \delta_g^{\overline{MS}} \mu^{\epsilon/2} = 0$$

$$\Rightarrow \delta_g^{\overline{MS}} \mu^{\epsilon/2} + \mu^{\epsilon/2} \frac{g^3}{2(4\pi)^3} \frac{2}{\epsilon} = 0$$

$$\Rightarrow \delta_g^{\overline{MS}} = - \frac{g^3}{2(4\pi)^3} \frac{2}{\epsilon} \quad \blacksquare$$

Exercise 7.6

The 1-loop diagram for the 1-point function is



← the momentum in loop is fixed by momentum conservation

The loop integral is $\mathcal{I} = \int \frac{d^d k}{(2\pi)^d} \frac{i}{(k^2 - m^2 + i\epsilon)}$

Vertex gives $-ig\mu^{\epsilon/2}$, the counter term gives $-i\delta_1$ (straight from Lagrangian)

1-point function is given by $1 + \Pi_1^{(1)}(p^2) - i\delta_1$ where $\Pi_1^{(1)}(p^2) = -ig\mu^{\epsilon/2} \mathcal{I}$

$$\mathcal{I} = \int \frac{d^d k}{(2\pi)^d} \frac{i}{k^2 - m^2 + i\epsilon}$$

Let's do Wick rotation: $k^0 \rightarrow i k_E^0$ and $k^i \rightarrow k_E^i$
 $\Rightarrow d^d k \rightarrow i d^d k_E$ and $k^2 \rightarrow -k_E^2$

$$\mathcal{I} = -i^2 \int \frac{d^d k_E}{(2\pi)^d} \frac{1}{(-k_E^2 - m^2)^1}$$

|| We can drop $i\epsilon$ since there is no singularity in the Euclidean integral.

$$\mathcal{I} = -i^2 \frac{1}{(4\pi)^{d/2}} \frac{\Gamma(1 - \frac{d}{2})}{\Gamma(1)} (m^2)^{\frac{d}{2} - 1}$$

|| Here we used the integration formulae from Peskin (7.85)

$$\mathcal{I} = + \frac{(4\pi)^{\epsilon/2}}{(4\pi)^3} \frac{\Gamma(\frac{\epsilon}{2} - 2)}{1} (m^2)^{2 - \epsilon/2}$$

$$\left\langle \begin{aligned} d &= 6 - \epsilon, & \Gamma(1) &= 1, & 1 - \frac{d}{2} &= -2 + \frac{\epsilon}{2} \end{aligned} \right.$$

$$\Gamma\left(\left(\frac{\epsilon}{2} - 1\right) - 1\right) = \left(\frac{\epsilon}{2} - 2\right)^{-1} \Gamma\left(\frac{\epsilon}{2} - 1\right)$$

$$\parallel \Gamma\left(\frac{\epsilon}{2} - 1\right) = \left(\frac{\epsilon}{2} - 1\right)^{-1} \Gamma\left(\frac{\epsilon}{2}\right)$$

$$\Gamma\left(\frac{\epsilon}{2} - 2\right) = \left(\frac{\epsilon}{2} - 2\right)^{-1} \left(\frac{\epsilon}{2} - 1\right)^{-1} \Gamma\left(\frac{\epsilon}{2}\right)$$

|| $(\epsilon - 2)^{-1} \approx -\frac{1}{2}$ and $(\epsilon - 1)^{-1} \approx -1$ are valid approximations since $\epsilon \ll 1$

$$\Gamma\left(\frac{\epsilon}{2} - 2\right) \approx +\frac{1}{2} \Gamma\left(\frac{\epsilon}{2}\right)$$

And no need to use binomial approx since at the end we drop terms of $\mathcal{O}(\epsilon)$ anyway

$$\Rightarrow I = + \frac{(4\pi)^{\epsilon/2}}{2(4\pi)^3} \Gamma(\frac{\epsilon}{2}) m^4 m^{-\epsilon}$$

$$\Rightarrow -i g \mu^{\epsilon/2} I = -i \frac{g(4\pi)^{\epsilon/2} \mu^{\epsilon/2}}{2(4\pi)^3} m^4 \Gamma(\frac{\epsilon}{2}) m^{-\epsilon} = \Pi_1^{(1)}$$

We now approximate: $(4\pi\mu)^{\epsilon/2} \approx 1 + \frac{\epsilon}{2} \ln(4\pi\mu)$ and $m^{-\epsilon} \approx 1 - \epsilon \ln(m)$
and $\Gamma(\frac{\epsilon}{2}) \approx \frac{2}{\epsilon} + \gamma_E$

$$\Rightarrow \Pi_{loop} = \Pi_1^{(1)} = -i \frac{g m^4}{2(4\pi)^3} \left(\frac{2}{\epsilon} + \gamma_E \right) \left(1 + \frac{\epsilon}{2} \ln(4\pi\mu) \right) (1 - \epsilon \ln(m))$$

$\epsilon \rightarrow 0$ so terms of order $\mathcal{O}(\epsilon)$ or higher vanish

$$\Pi_{loop} \approx -i \frac{g m^4}{2(4\pi)^3} \left(\frac{2}{\epsilon} + \gamma_E + \ln(4\pi\mu) - 2\ln(m) \right)$$

The divergent part has $\frac{2}{\epsilon}$ and is the only term that needs to be countered by δ_1 in $\overline{MS}$ -scheme.

$$\Rightarrow -i \frac{g m^4}{2(4\pi)^3} \frac{2}{\epsilon} - i \delta_1^{MS} = 0$$

$$\Rightarrow \underline{\delta_1^{MS} = - \frac{g m^4}{2(4\pi)^3} \frac{2}{\epsilon}} \quad \square$$

This tadpole amplitude can be written as $\delta_1^{MS} = m^2 \delta_m^{MS}$, so there is connection with δ_m^{MS} .

If we allow the loops of 2-point function to also contain tadpole diagrams then there is no need for δ_1 -counter term i.e. we allow Feynman diagram:



to be allowed as 1-loop correction to

the 2-point function. $\Rightarrow$ No need for distinct δ_1 -counterterm.

For higher order corrections we allow tadpole contributions to similarly be in the 2-point function corrections.